Answer **all** the questions in this section.

1 **(a)** Fig 1.1 shows two identical beakers. A wooden block is placed inside beaker B. In both beakers, the water level is filled to the brim. The two beakers are each placed on a top-pan balance.

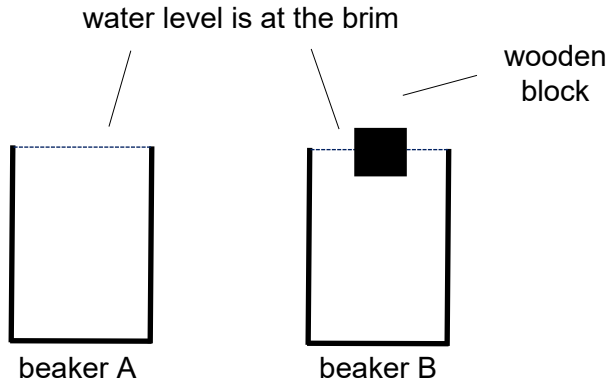

Fig. 1.1

Compare the top-pan balance readings for the two beakers.

Explain your answer.

.....

.....

.....

..... [2]

(b) Fig.1.2 shows a beaker of water placed on a force sensor connected to a data-logger. A solid block is submerged in the water. A string is attached on one end to the ceiling, and the other end to the solid block.

The beaker and water have combined weight *P*. The solid block has weight *Q* and it displaces water of weight *R* when submerged. The tension in the string supporting the block is *T* and the contact force registered by the data-logger is *C*.

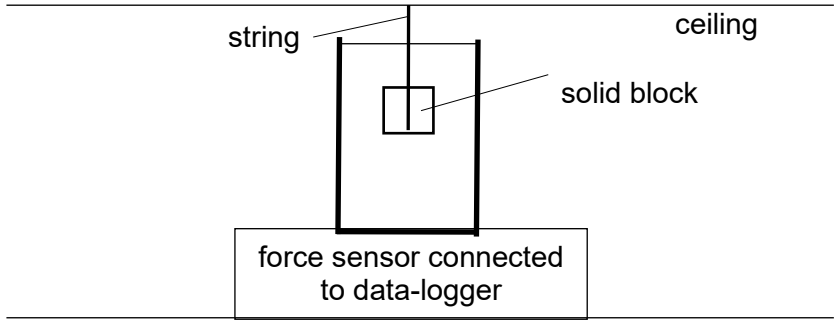

Fig. 1.2

(i) Draw a force diagram showing all the forces acting on

1. the solid block,

2. the system consisting of the solid block, beaker and water.
(You may treat the system as a point object.)

[3]

(ii) Fig 1.3 shows a table of values for *P*, *Q* and *R*.

combined weight of beaker and water, *P*	3.00 N
weight of solid block, *Q*	1.13 N
weight of water displaced by solid block, *R*	0.20 N

Fig. 1.3

1. Calculate the reading registered on the data-logger.

reading = ...N [2]

2. The string is now cut and the solid block falls and rests at the bottom of the beaker. Determine the change in the reading on the data-logger.

change in reading = ...N [2]